

Kafka Execution (Kafka version – 4.1.1)

(for new creation start from step 1 else proceed from step-3)

1. Create new Cluster ID

1.Open Command Prompt (cmd) and navigate to your Kafka directory:

```
cd C:\kafka
```

2.Run the following command to generate a unique cluster UUID

```
.\bin\windows\kafka-storage.sh random-uuid
```

3.Copy the generated UUID string that appears in your console

2. Format Log Directories

1.Use the copied UUID to format the **directories (replace <your-cluster-id> with the UUID you copied)**:

```
bin\windows\kafka-storage.sh format --standalone -t <your-cluster-id> -c config\kraft\server.properties
```

2.Copy the generated UUID from the output and use it to replace <your-uuid-here> in the following command

```
.\bin\windows\kafka-storage.sh format --standalone -t <your-uuid-here> -c .\config\server.properties
```

3. Start the Kafka Server (applicable for both new/ existing kafka server)

1. Ensure the kafka post is uncommented the **server.properties** file in config & under windows folder of bin

```
listeners=PLAINTEXT://:9092
```

Note: edit the port number in above file if you need to change the port number

2.Check the below format also in same server.properties file and modify the port number which you have given in above place only for listener.

```
listeners=PLAINTEXT://localhost:9092,CONTROLLER://localhost:9093
```

Note:

PLAINTEXT://localhost:9092 (Client Traffic)

- **Purpose:** This is the port where external applications (like your **Spring Boot** apps, producers, or consumers) connect to send and receive data.
- **Protocol:** Unencrypted data (PLAINTEXT).

CONTROLLER://localhost:9093 (Cluster Traffic)

- **Purpose:** This is the port used internally by Kafka nodes to communicate with each other to manage cluster metadata, replacing the old ZooKeeper communication.
- **Protocol:** Internal metadata synchronization.

2.Run the below command to start the kafka server with port number mentioned in above step for starting kafka server

bin\windows\kafka-server-start.bat config\server.properties

```
S:\SpringBoot\Apache_Kafka\kafka_2.13-4.1.1>bin\windows\kafka-server-start.bat config\server.properties
DEPRECATED: A Log4j 1.x configuration file has been detected, which is no longer recommended.
To use a Log4j 2.x configuration, please see https://logging.apache.org/log4j/2.x/migrate-from-log4j1.html#Log4j2ConfigurationFormat for details about Log4j
configuration file migration.
You can also use the S:\SpringBoot\Apache_Kafka\kafka_2.13-4.1.1\config\tool-log4j2.yaml file as a starting point. Make sure to remove the Log4j 1.x configu
ration after completing the migration.
[2026-05-27 19:45:54,912] INFO Registered 'kafka:type=kafka.Log4jController' MBean (kafka.utils.Log4jControllerRegistration$)
[2026-05-27 19:45:55,399] INFO [ControllerServer id=1] Starting controller (kafka.server.ControllerServer)
[2026-05-27 19:45:56,076] INFO Updated connection-accept-rate max connection creation rate to 2197083607 (kafka.network.ConnectionQuotas)
[2026-05-27 19:45:56,187] INFO [SocketServer listenerType=CONTROLLER, nodeId=1] Created data-plane acceptor and processors for endpoint : ListenerName(CONTR
OLLER) (kafka.network.SocketServer)
[2026-05-27 19:45:56,202] INFO authorizerStart completed for endpoint CONTROLLER. Endpoint is now READY. (org.apache.kafka.server.network.EndpointReadyFutur
es)
[2026-05-27 19:45:56,202] INFO [SharedServer id=1] Starting SharedServer (kafka.server.SharedServer)
[2026-05-27 19:45:56,313] INFO [LogLoader partition=__cluster_metadata-0, dir=S:\tmp\kraft-combined-logs] Recovering unflushed segment 0. 0 recovered for __
cluster_metadata-0. (org.apache.kafka.storage.internals.log.LogLoader)
[2026-05-27 19:45:56,313] INFO [LogLoader partition=__cluster_metadata-0, dir=S:\tmp\kraft-combined-logs] Loading producer state till offset 0 (org.apache.k
afka.storage.internals.log.UnifiedLog)
[2026-05-27 19:45:56,313] INFO [LogLoader partition=__cluster_metadata-0, dir=S:\tmp\kraft-combined-logs] Reloading from producer snapshot and rebuilding pr
oducer state from offset 0 (org.apache.kafka.storage.internals.log.UnifiedLog)
[2026-05-27 19:45:56,319] INFO Deleted producer state snapshot S:\tmp\kraft-combined-logs\__cluster_metadata-0\00000000000000000549.snapshot (org.apache.kaf
ka.storage.internals.log.SnapshotFile)
[2026-05-27 19:45:56,319] INFO Deleted producer state snapshot S:\tmp\kraft-combined-logs\__cluster_metadata-0\00000000000000014753.snapshot (org.apache.kaf
ka.storage.internals.log.SnapshotFile)
[2026-05-27 19:45:56,319] INFO [LogLoader partition=__cluster_metadata-0, dir=S:\tmp\kraft-combined-logs] Producer state recovery took 6ms for snapshot load
and 0ms for segment recovery from offset 0 (org.apache.kafka.storage.internals.log.UnifiedLog)
[2026-05-27 19:45:56,534] INFO [ProducerStateManager partition=__cluster_metadata-0] Wrote producer snapshot at offset 14753 with 0 producer ids in 6 ms. (o
rg.apache.kafka.storage.internals.log.ProducerStateManager)
[2026-05-27 19:45:56,551] INFO [LogLoader partition=__cluster_metadata-0, dir=S:\tmp\kraft-combined-logs] Loading producer state till offset 14753 (org.apac
he.kafka.storage.internals.log.UnifiedLog)
[2026-05-27 19:45:56,554] INFO [LogLoader partition=__cluster_metadata-0, dir=S:\tmp\kraft-combined-logs] Reloading from producer snapshot and rebuilding pr
oducer state from offset 14753 (org.apache.kafka.storage.internals.log.UnifiedLog)
[2026-05-27 19:45:56,559] INFO [ProducerStateManager partition=__cluster_metadata-0] Loading producer state from snapshot file 'SnapshotFile(offset=14753, f
ile=S:\tmp\kraft-combined-logs\__cluster_metadata-0\00000000000000014753.snapshot)' (org.apache.kafka.storage.internals.log.ProducerStateManager)
[2026-05-27 19:45:56,586] INFO [LogLoader partition=__cluster_metadata-0, dir=S:\tmp\kraft-combined-logs] Producer state recovery took 32ms for snapshot loa
```

```
[2026-05-27 19:45:59,760] INFO [GroupCoordinator id=1] Finished loading of metadata from __consumer_offsets-30 with epoch 2 in 4ms where 4ms was spent in th
e scheduler. Loaded 0 records which total to 0 bytes. (org.apache.kafka.coordinator.common.runtime.CoordinatorRuntime)
[2026-05-27 19:45:59,790] INFO [GroupCoordinator id=1] Finished loading of metadata from __consumer_offsets-33 with epoch 2 in 59ms where 59ms was spent in
the scheduler. Loaded 0 records which total to 0 bytes. (org.apache.kafka.coordinator.common.runtime.CoordinatorRuntime)
[2026-05-27 19:45:59,790] INFO [GroupCoordinator id=1 topic=__consumer_offsets partition=8] Loaded classic group Kafka_listener with 1 members. (org.apache.k
afka.coordinator.group.GroupMetadataManager)
[2026-05-27 19:45:59,791] INFO [GroupCoordinator id=1] Finished loading of metadata from __consumer_offsets-8 with epoch 2 in 61ms where 3ms was spent in th
e scheduler. Loaded 46 records which total to 10684 bytes. (org.apache.kafka.coordinator.common.runtime.CoordinatorRuntime)
[2026-05-27 19:46:00,798] INFO [GroupCoordinator id=1 topic=__consumer_offsets partition=8] Member consumer-Kafka_listener-1-1d604ba2-6b7a-4b32-b5f2-55b2a6e4
d590 in group Kafka_listener has failed, removing it from the group. (org.apache.kafka.coordinator.group.GroupMetadataManager)
[2026-05-27 19:46:00,798] INFO [GroupCoordinator id=1 topic=__consumer_offsets partition=8] Preparing to rebalance group Kafka_listener in state PreparingReb
alance with old generation 43 (reason: removing member consumer-Kafka_listener-1-1d604ba2-6b7a-4b32-b5f2-55b2a6e4d590 on heartbeat expiration.). (org.apache
.kafka.coordinator.group.GroupMetadataManager)
[2026-05-27 19:46:00,809] INFO [GroupCoordinator id=1 topic=__consumer_offsets partition=8] Group Kafka_listener with generation 44 is now empty. (org.apache
.kafka.coordinator.group.GroupMetadataManager)
[2026-05-27 19:55:57,858] INFO [NodeToControllerChannelManager id=1 name=registration] Node 1 disconnected. (org.apache.kafka.clients.NetworkClient)
[2026-05-27 20:00:27,147] INFO [NodeToControllerChannelManager id=1 name=forwarding] Node 1 disconnected. (org.apache.kafka.clients.NetworkClient)
[2026-05-27 20:00:33,520] INFO [GroupCoordinator id=1 topic=__consumer_offsets partition=8] Dynamic member with unknown member id joins group Kafka_listener
in Empty state. Created a new member id consumer-Kafka_listener-1-7c7ed03d-4698-4dd3-b9ae-e4a68a8b39e5 and requesting the member to rejoin with this id. (org
.apache.kafka.coordinator.group.GroupMetadataManager)
[2026-05-27 20:09:33,501] INFO [GroupCoordinator id=1 topic=__consumer_offsets partition=8] Pending dynamic member with id consumer-Kafka_listener-1-7c7ed03d-
4698-4dd3-b9ae-e4a68a8b39e5 joins group Kafka_listener in Empty state. Adding to the group now. (org.apache.kafka.coordinator.group.GroupMetadataManager)
[2026-05-27 20:09:33,551] INFO [GroupCoordinator id=1 topic=__consumer_offsets partition=8] Preparing to rebalance group Kafka_listener in state PreparingReb
alance with old generation 44 (reason: Adding new member consumer-Kafka_listener-1-7c7ed03d-4698-4dd3-b9ae-e4a68a8b39e5 with group instance id null; client r
eason: need to re-join with the given member-id: consumer-Kafka_listener-1-7c7ed03d-4698-4dd3-b9ae-e4a68a8b39e5). (org.apache.kafka.coordinator.group.GroupMe
tadataManager)
[2026-05-27 20:09:36,564] INFO [GroupCoordinator id=1 topic=__consumer_offsets partition=8] Stabilized group Kafka_listener generation 45 with 1 members. (or
g.apache.kafka.coordinator.group.GroupMetadataManager)
[2026-05-27 20:09:36,611] INFO [GroupCoordinator id=1 topic=__consumer_offsets partition=8] Assignment received from leader consumer-Kafka_listener-1-7c7ed03
d-4698-4dd3-b9ae-e4a68a8b39e5 for group Kafka_listener for generation 45. The group has 1 members, 0 of which are static. (org.apache.kafka.coordinator.group
.GroupMetadataManager)
[2026-05-27 20:18:53,412] INFO [GroupCoordinator id=1 topic=__consumer_offsets partition=8] [Group Kafka_listener] Member consumer-Kafka_listener-1-7c7ed03d-4
698-4dd3-b9ae-e4a68a8b39e5 has left group through explicit 'LeaveGroup' request; client reason: the consumer unsubscribed from all topics (org.apache.kafka
.coordinator.group.GroupMetadataManager)
[2026-05-27 20:18:53,412] INFO [GroupCoordinator id=1 topic=__consumer_offsets partition=8] Preparing to rebalance group Kafka_listener in state PreparingReb
alance with old generation 45 (reason: explicit 'LeaveGroup' request for (consumer-Kafka_listener-1-7c7ed03d-4698-4dd3-b9ae-e4a68a8b39e5) members.). (org.apa
che.kafka.coordinator.group.GroupMetadataManager)
[2026-05-27 20:18:53,412] INFO [GroupCoordinator id=1 topic=__consumer_offsets partition=8] Group Kafka_listener with generation 46 is now empty. (org.apache
.kafka.coordinator.group.GroupMetadataManager)
[2026-05-27 20:19:42,376] INFO [NodeToControllerChannelManager id=1 name=forwarding] Node 1 disconnected. (org.apache.kafka.clients.NetworkClient)
```

4.Create new topic: (Not required everytime, If you want to use existing topic skip this step)

Run below command to create a new topic

```
bin\windows\kafka-topics.bat --create --topic <my-first-topic> --bootstrap-server localhost:<portnumber where server is started> --partitions 1 --replication-factor 1
```

5. Check created or existing Topics:

```
bin\windows\Kafka-topics.bat --describe --topic <topic-name> --bootstrap-server localhost:<portnumber where server is started>
```

6. List all topics created:

```
bin\windows\Kafka-topics.bat --list --bootstrap-server localhost:<portnumber where server is started>
```

```
Microsoft Windows [Version 10.0.22621.3807]
(c) Microsoft Corporation. All rights reserved.

C:\Users\iamsa>cd /d S:\SpringBoot\Apache_Kafka\kafka_2.13-4.1.1
The filename, directory name, or volume label syntax is incorrect.

C:\Users\iamsa>cd /d S:\SpringBoot\Apache_Kafka\kafka_2.13-4.1.1

S:\SpringBoot\Apache_Kafka\kafka_2.13-4.1.1>bin\windows\Wafka-topics.bat --describe --topic course --bootstrap-server localhost:9092
2026-05-27T14:23:59.949241200Z main ERROR Reconfiguration failed: No configuration found for '3cd1a2f1' at 'null' in 'null'
Error while executing topic command : Topic 'course' does not exist as expected
19:54:00.528 (main) ERROR org.apache.kafka.tools.TopicCommand - java.lang.IllegalArgumentException: Topic 'course' does not exist as expected
    at org.apache.kafka.tools.TopicCommand.ensureTopicExists(TopicCommand.java:211)
    at org.apache.kafka.tools.TopicCommand$TopicService.describeTopic(TopicCommand.java:571)
    at org.apache.kafka.tools.TopicCommand.execute(TopicCommand.java:109)
    at org.apache.kafka.tools.TopicCommand.mainNoExit(TopicCommand.java:88)
    at org.apache.kafka.tools.TopicCommand.main(TopicCommand.java:83)

S:\SpringBoot\Apache_Kafka\kafka_2.13-4.1.1>bin\windows\Wafka-topics.bat --describe --topic Course --bootstrap-server localhost:9092
2026-05-27T14:24:26.583617100Z main ERROR Reconfiguration failed: No configuration found for '3cd1a2f1' at 'null' in 'null'
Topic: Course TopicId: YjvckgNlQul25jgnPPXfQ PartitionCount: 1 ReplicationFactor: 1 Configs: min.insync.replicas=1,segment.bytes=1073741824
Topic: Course Partition: 0 Leader: 1 Replicas: 1 Isr: 1 Elr: LastKnownElr:

S:\SpringBoot\Apache_Kafka\kafka_2.13-4.1.1>bin\windows\Wafka-topics.bat --list --bootstrap-server localhost:9092
2026-05-27T14:29:21.154372100Z main ERROR Reconfiguration failed: No configuration found for '3cd1a2f1' at 'null' in 'null'
Course
...consumer_offsets
my-first-topic

S:\SpringBoot\Apache_Kafka\kafka_2.13-4.1.1>
```

Now you can run the application once after ensuring that the server is start and running